Distributed Quantum Computing

About Me

Three main research topics

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String Matching
(for graphs)

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Quantum Computing

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String Matching (for graphs)	Quantum Computing	Distributed Computing

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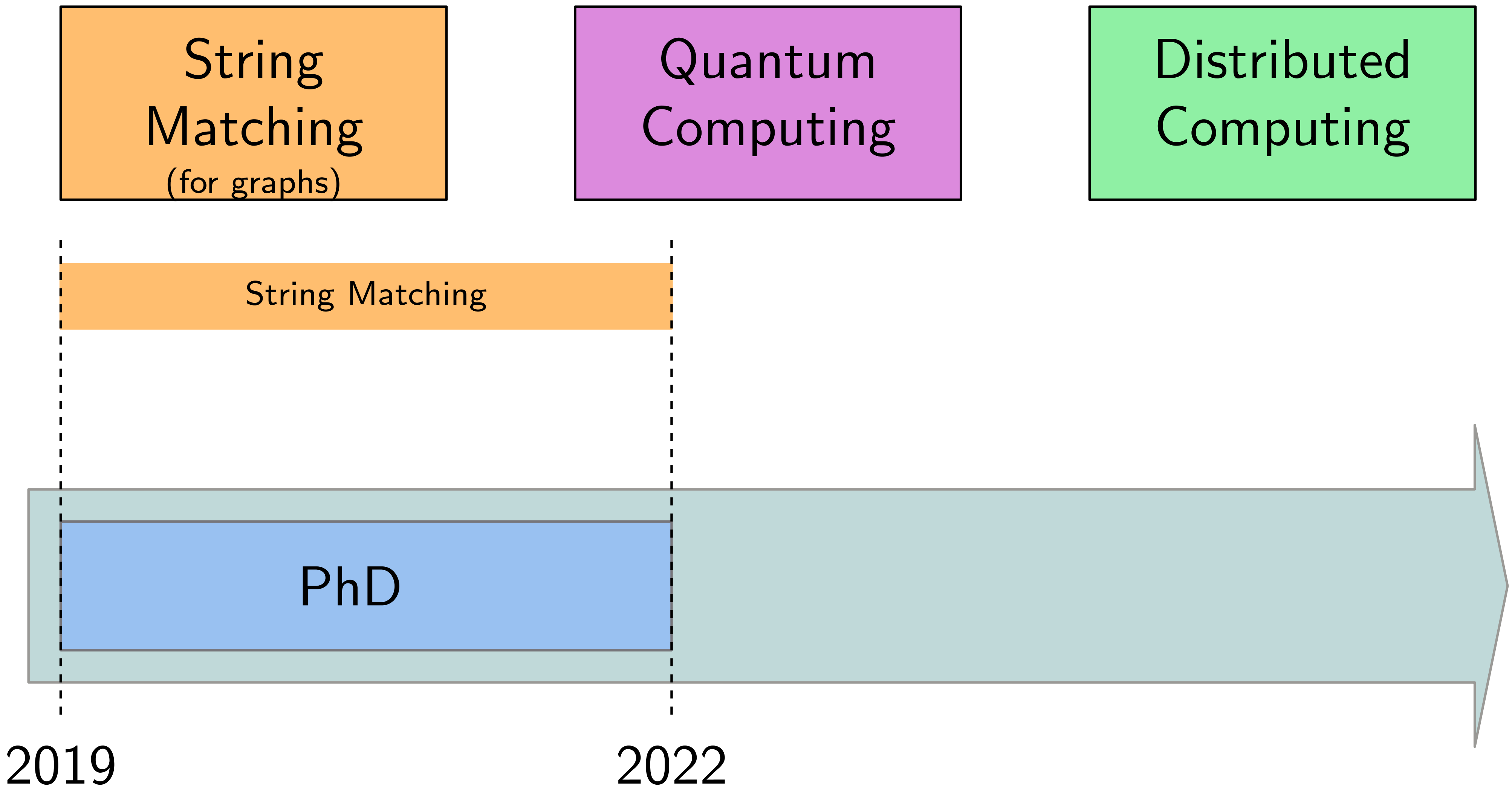

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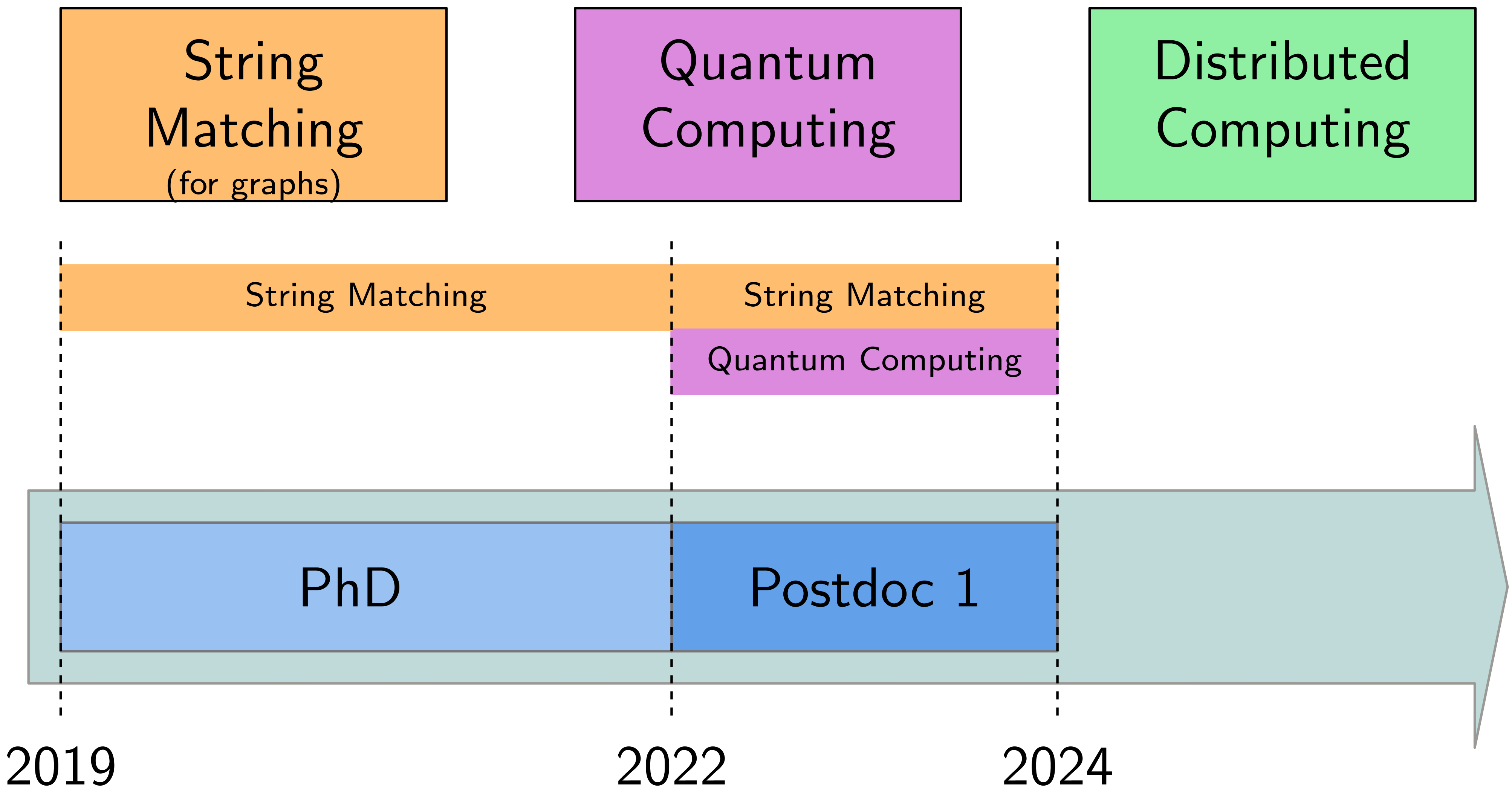

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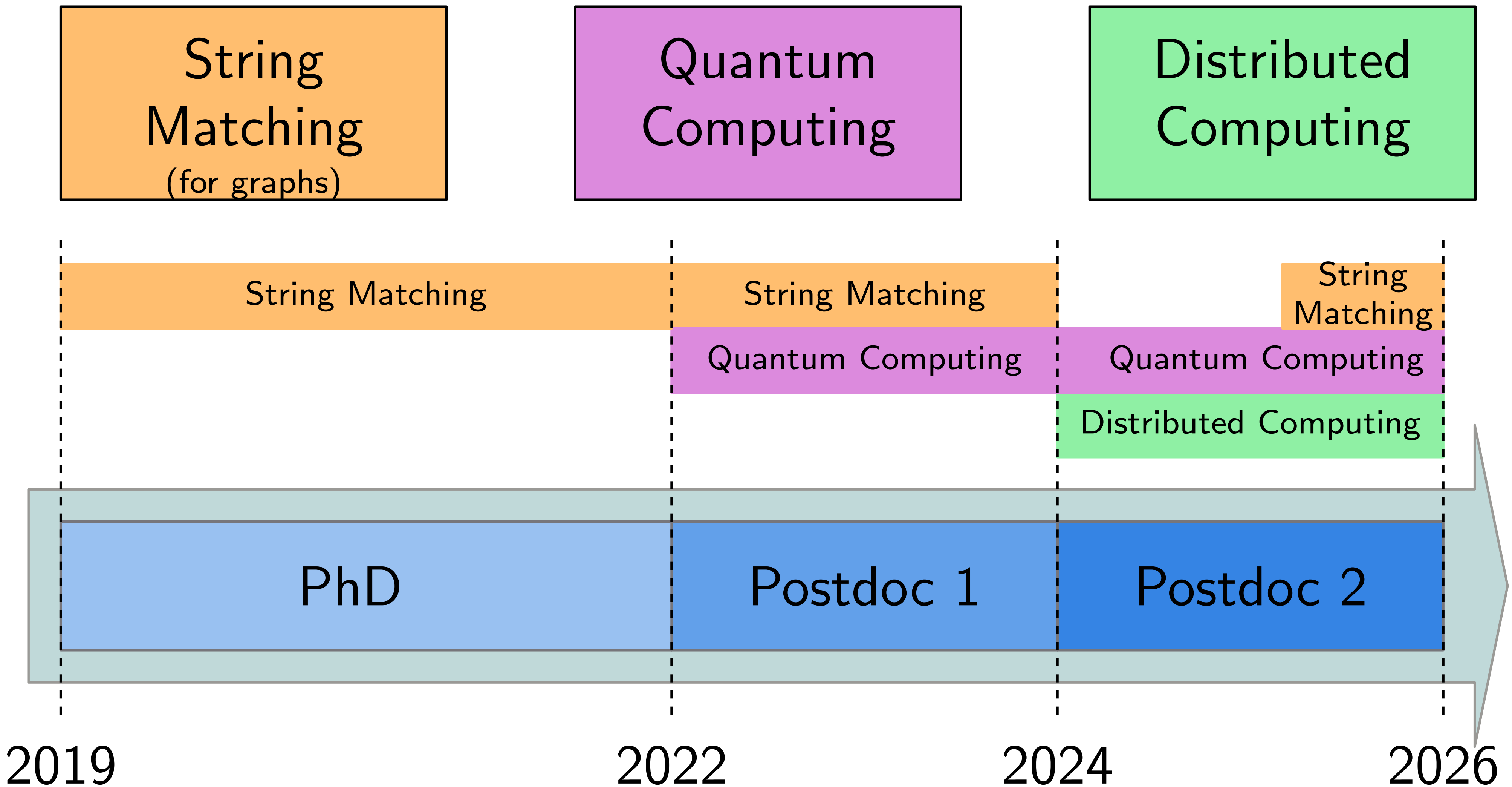
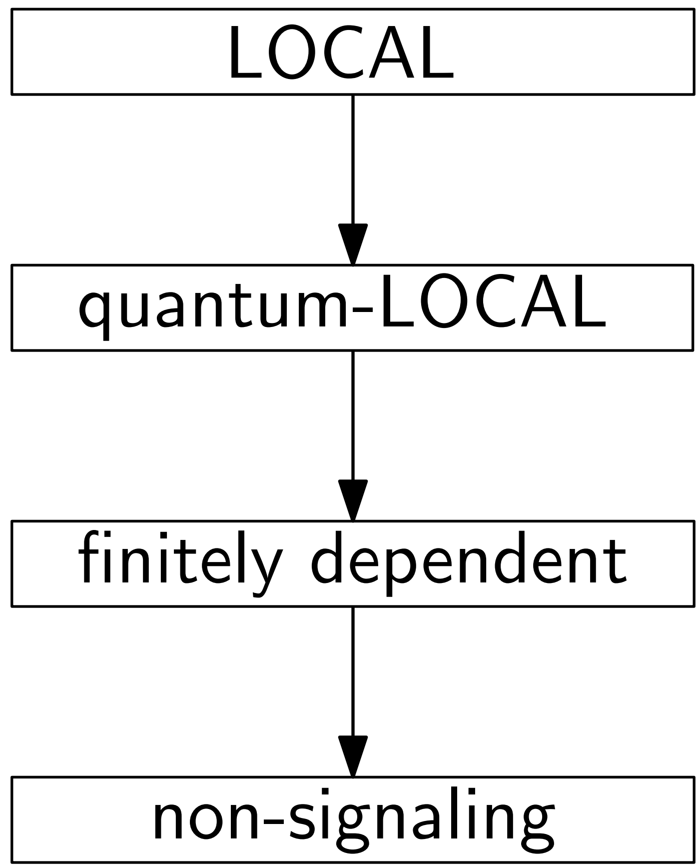
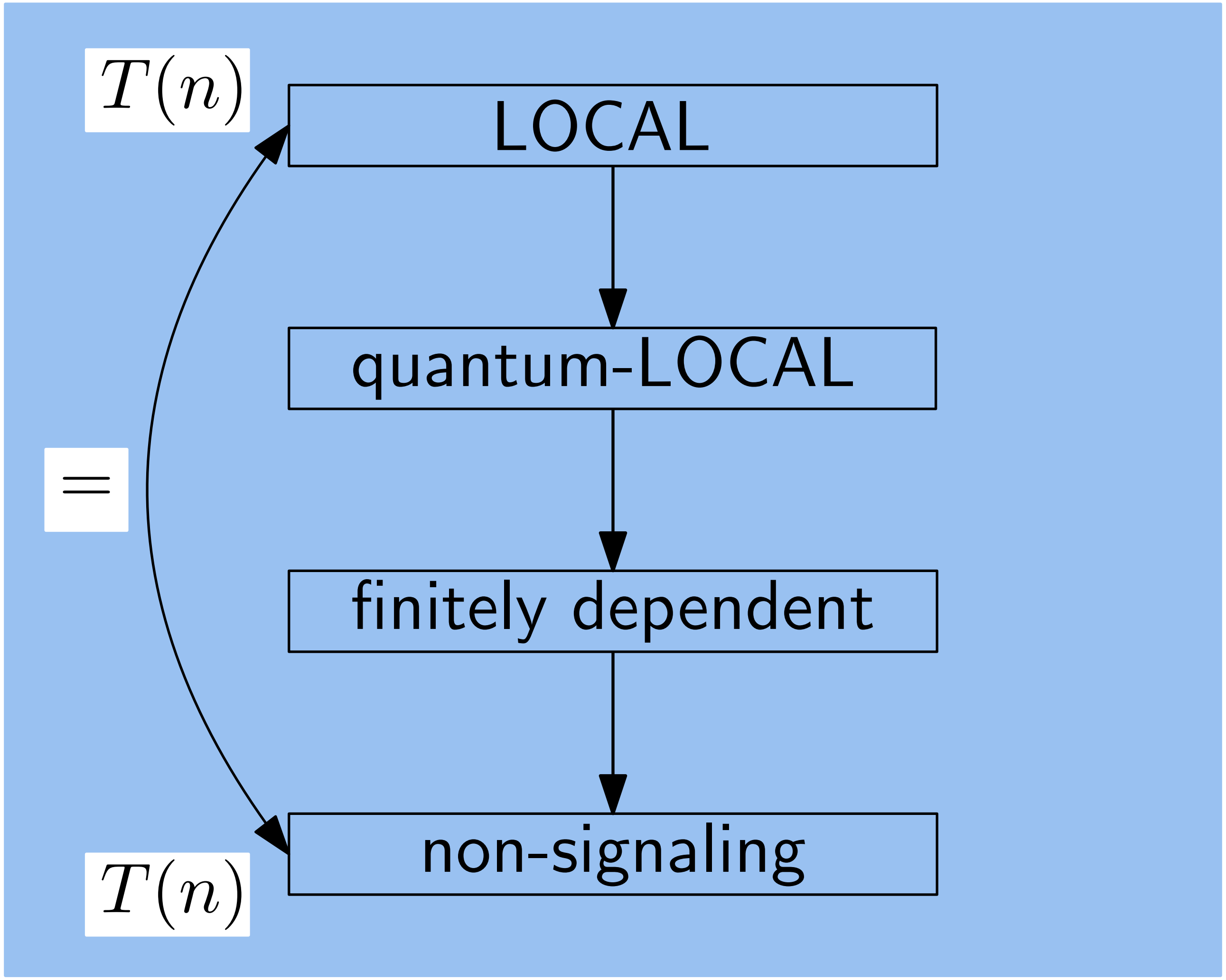


Models of Computations



Models of Computations

Linear Programs (LPs)



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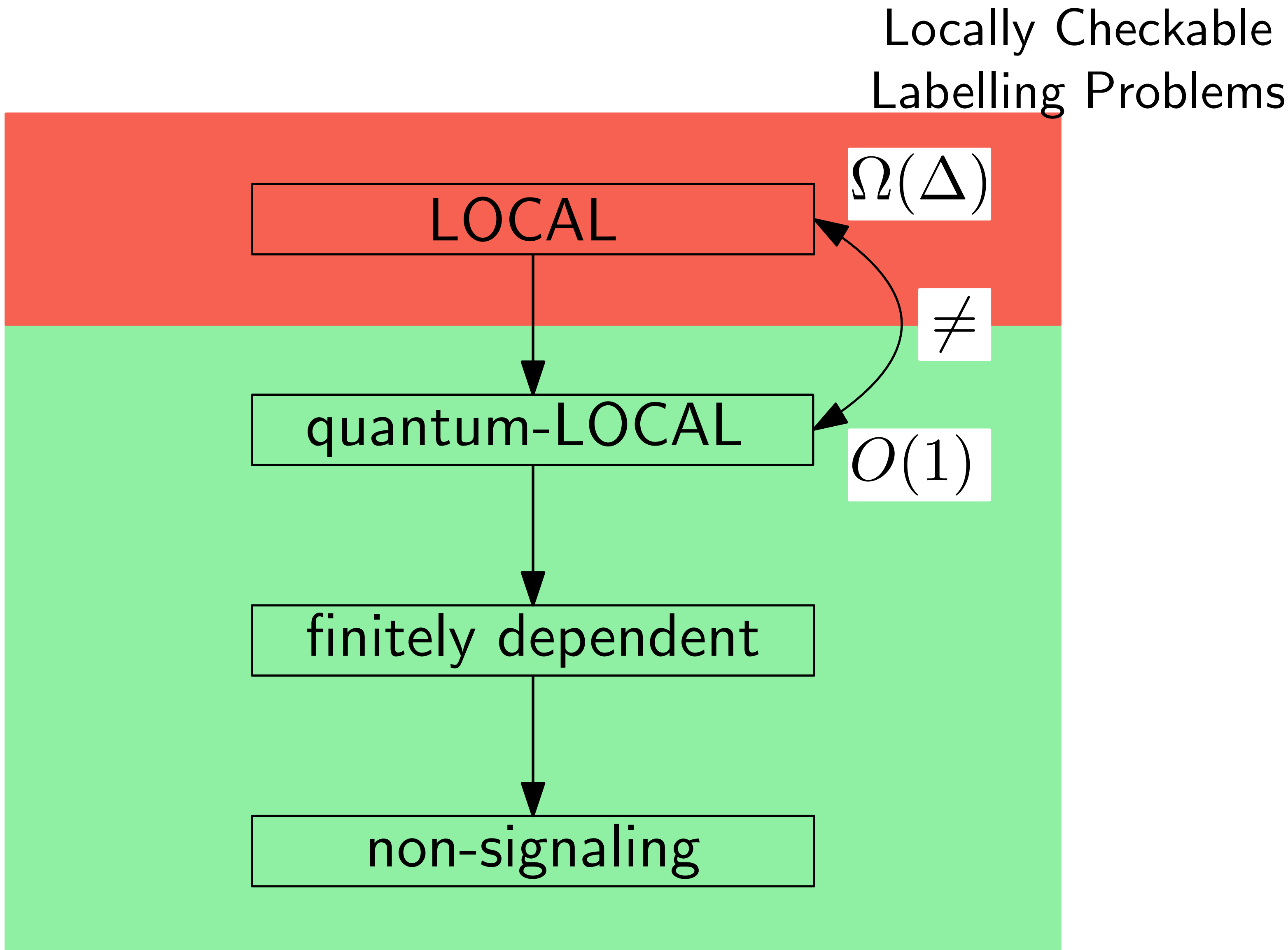

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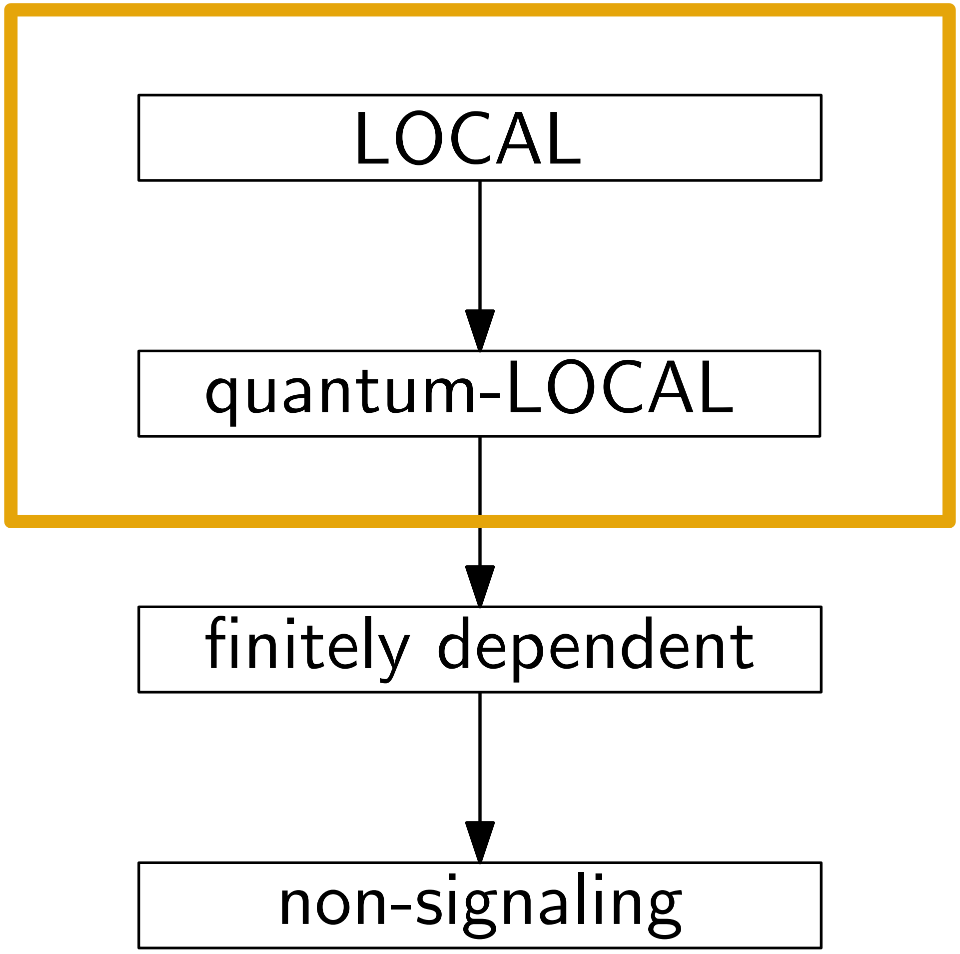
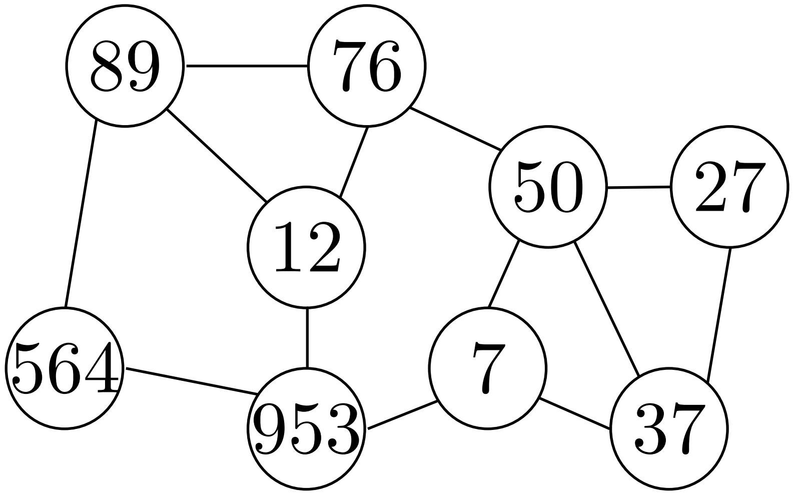


LOCAL and quantum-LOCAL



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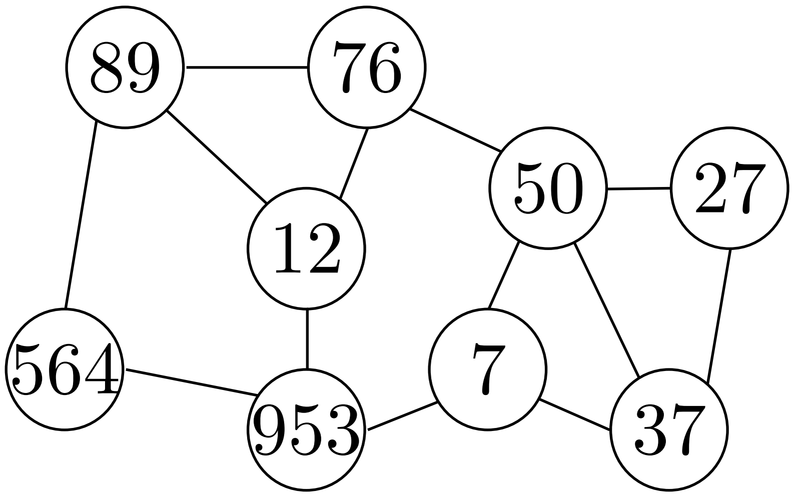

- The network is a graph $G = (V, E)$ of n nodes

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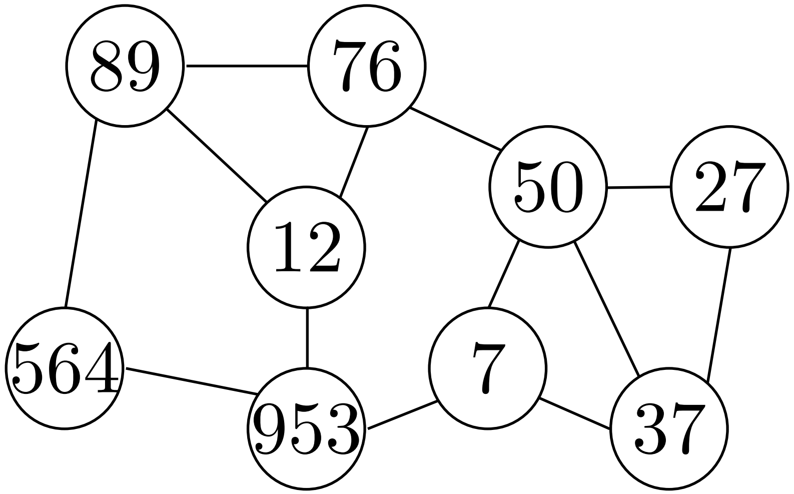

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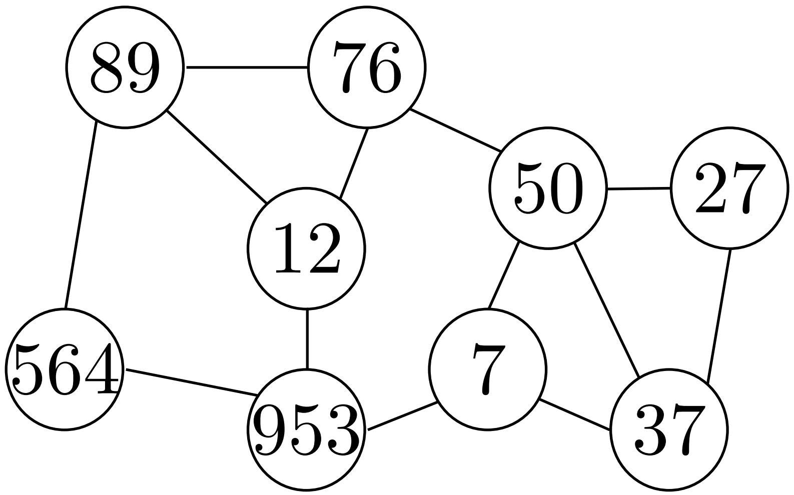

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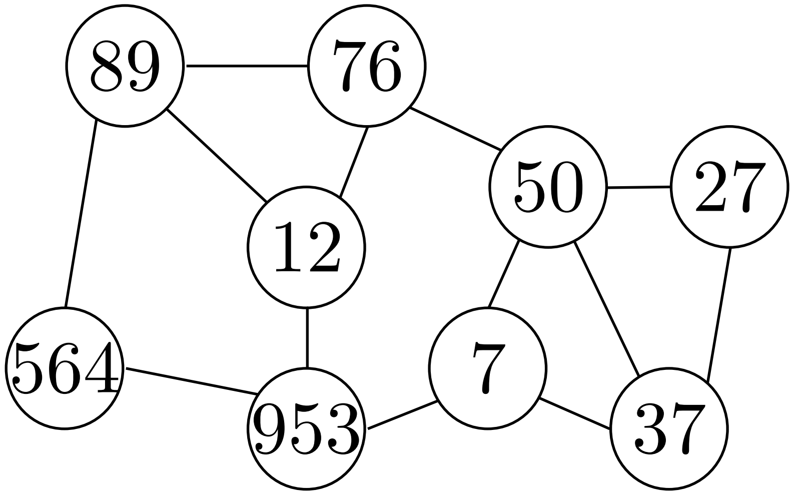

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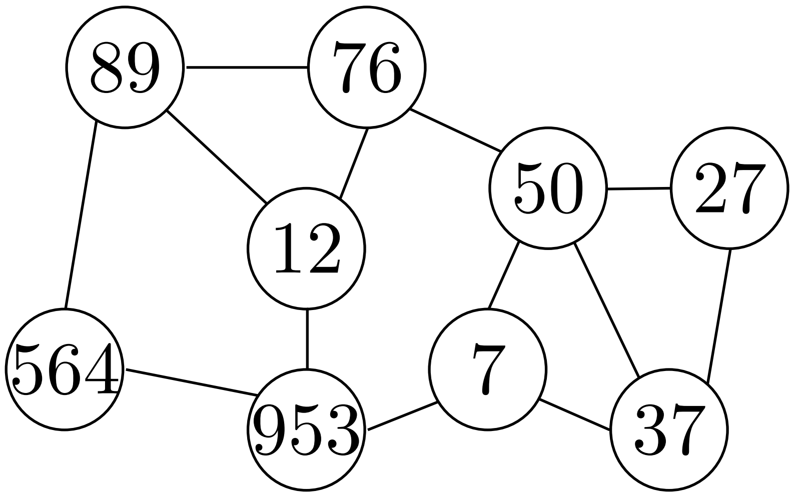

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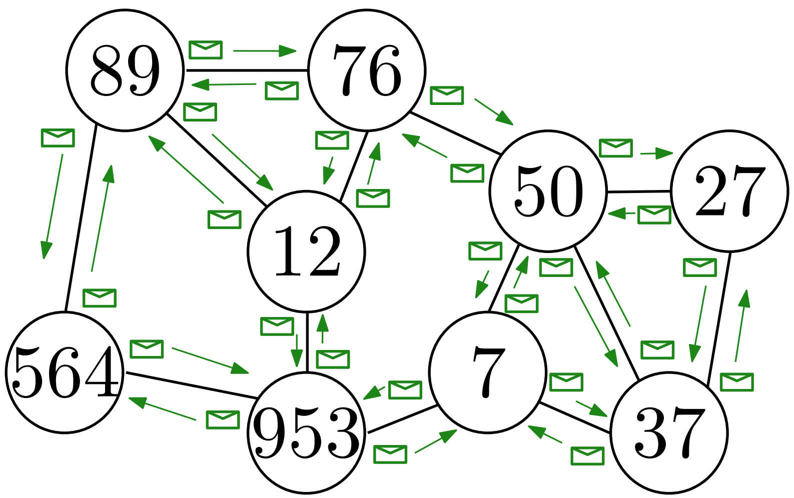

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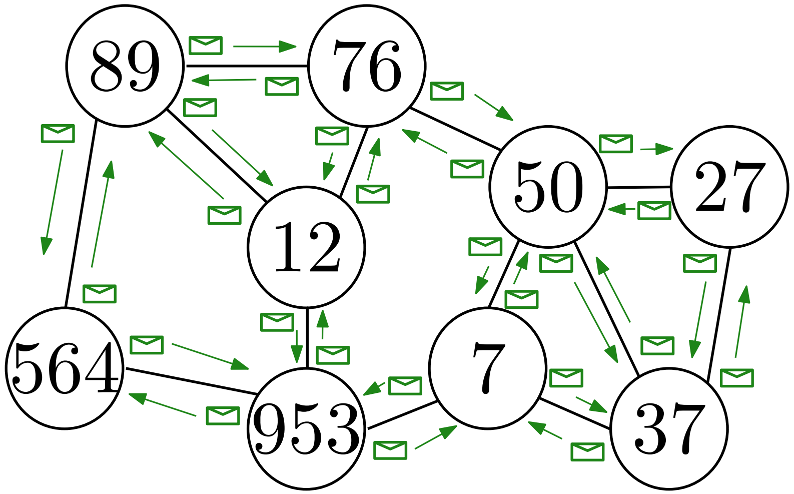
Complexity

Number of communication rounds as a function of n

LOCAL and quantum-LOCAL



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Nodes hold, send and receive **bits**

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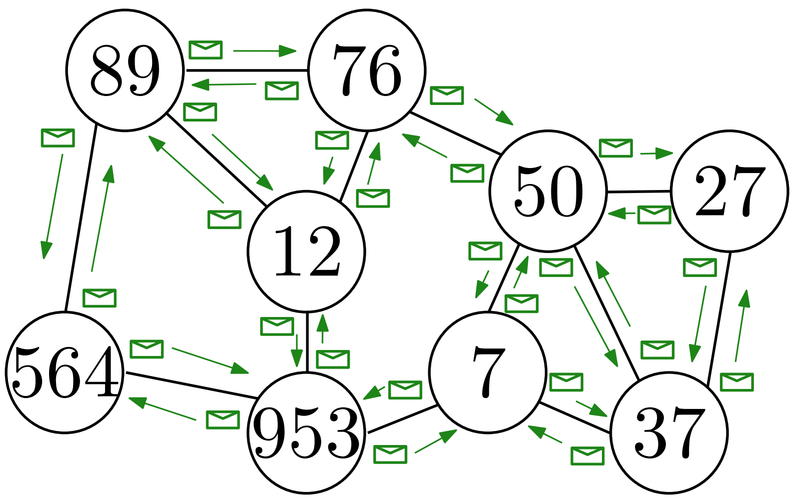

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Complexity

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LOCAL

Nodes hold, send and receive **bits**

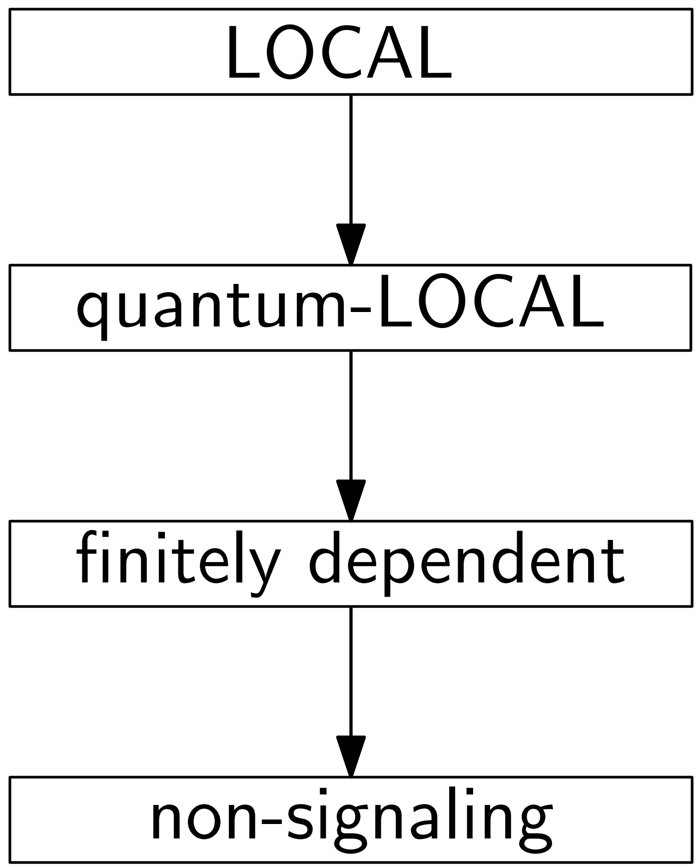
quantum-LOCAL

Nodes hold, send and receive **qubits**

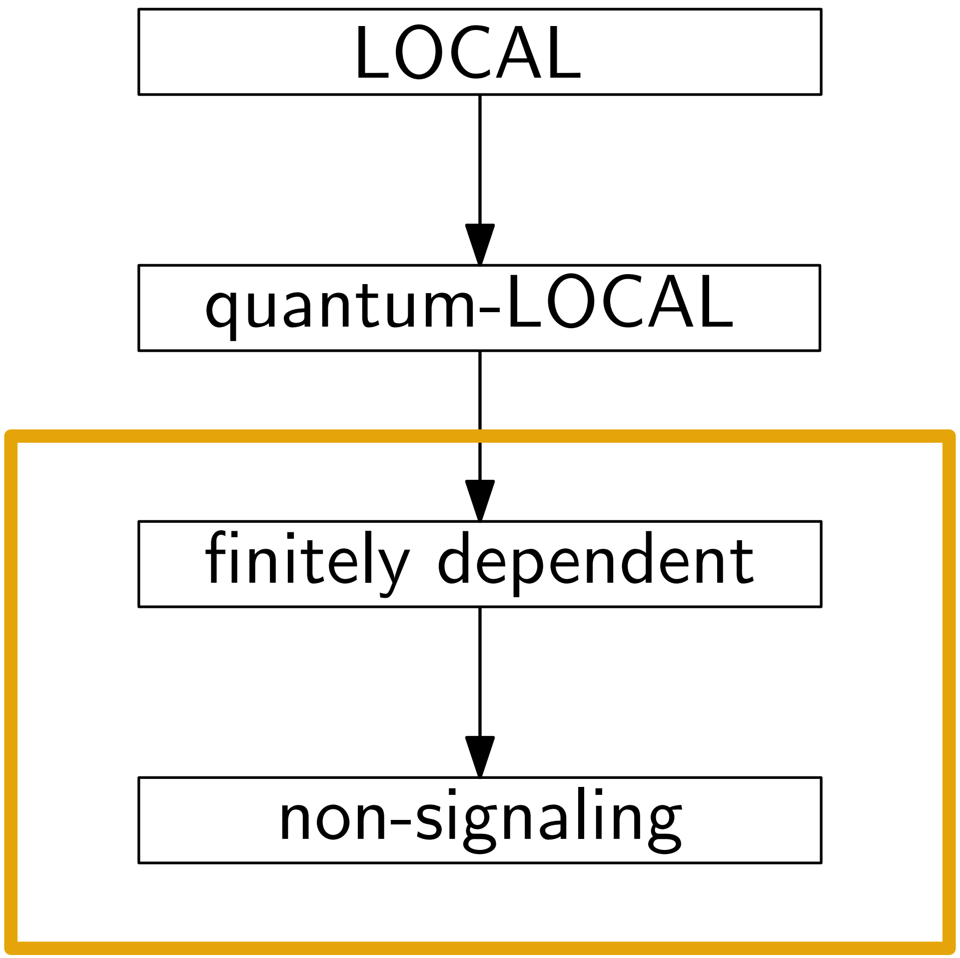
IDs are not needed

"send" qubits means applying SWAP gates

Models of Computations

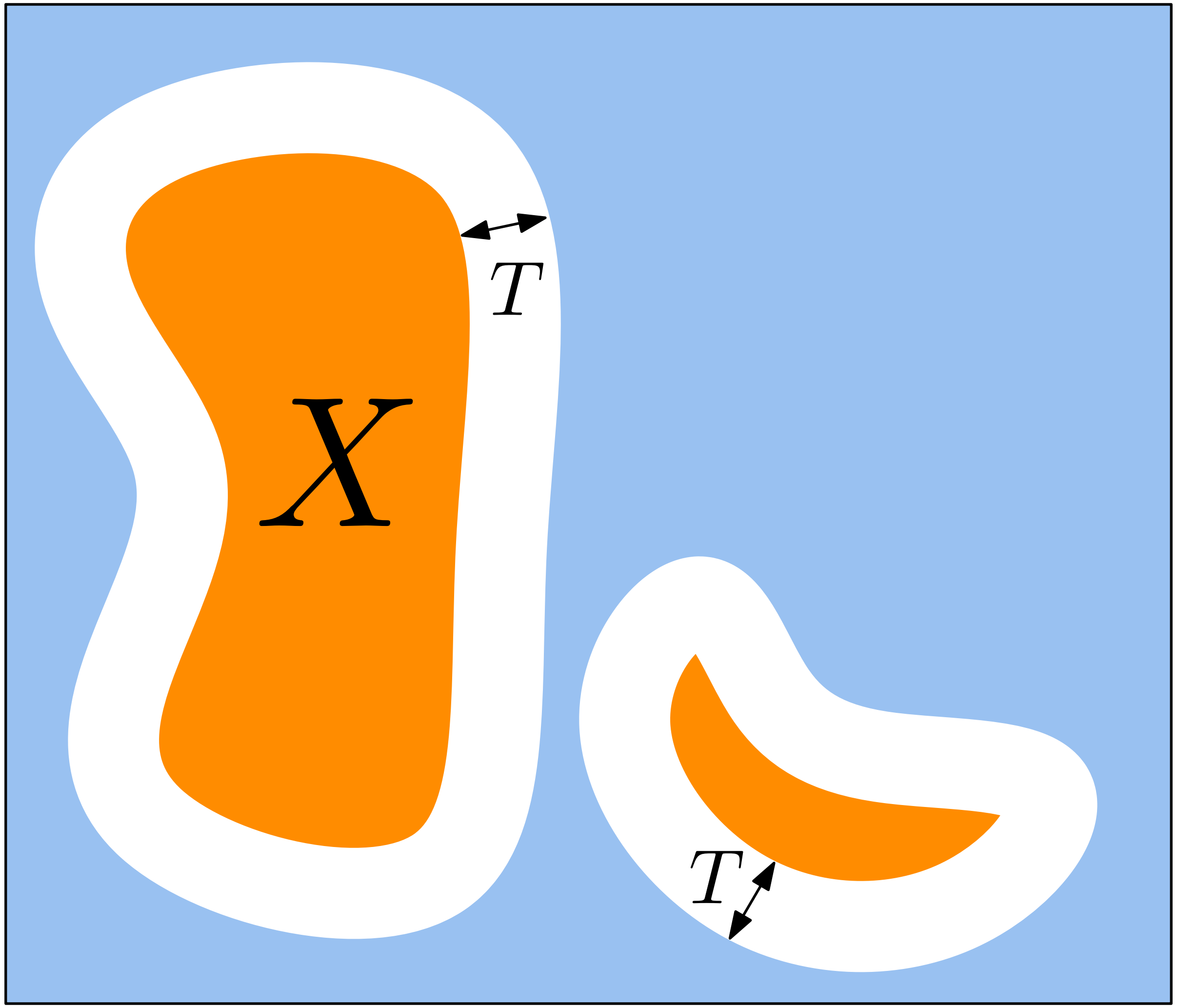


Models of Computations



Non-signaling distributions

For any subset of nodes X, changes at distance $> T$ **do not affect** the output distribution of X



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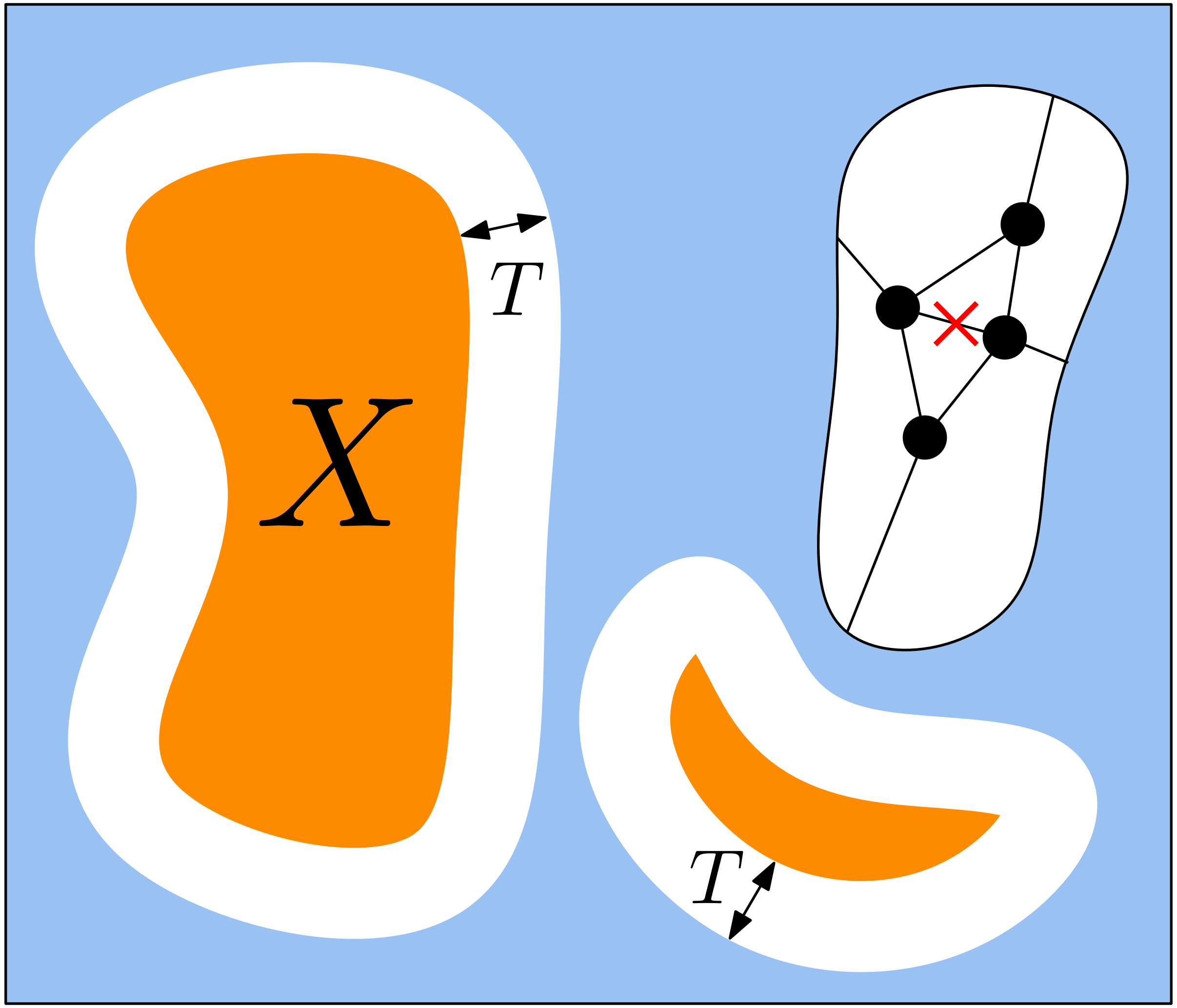

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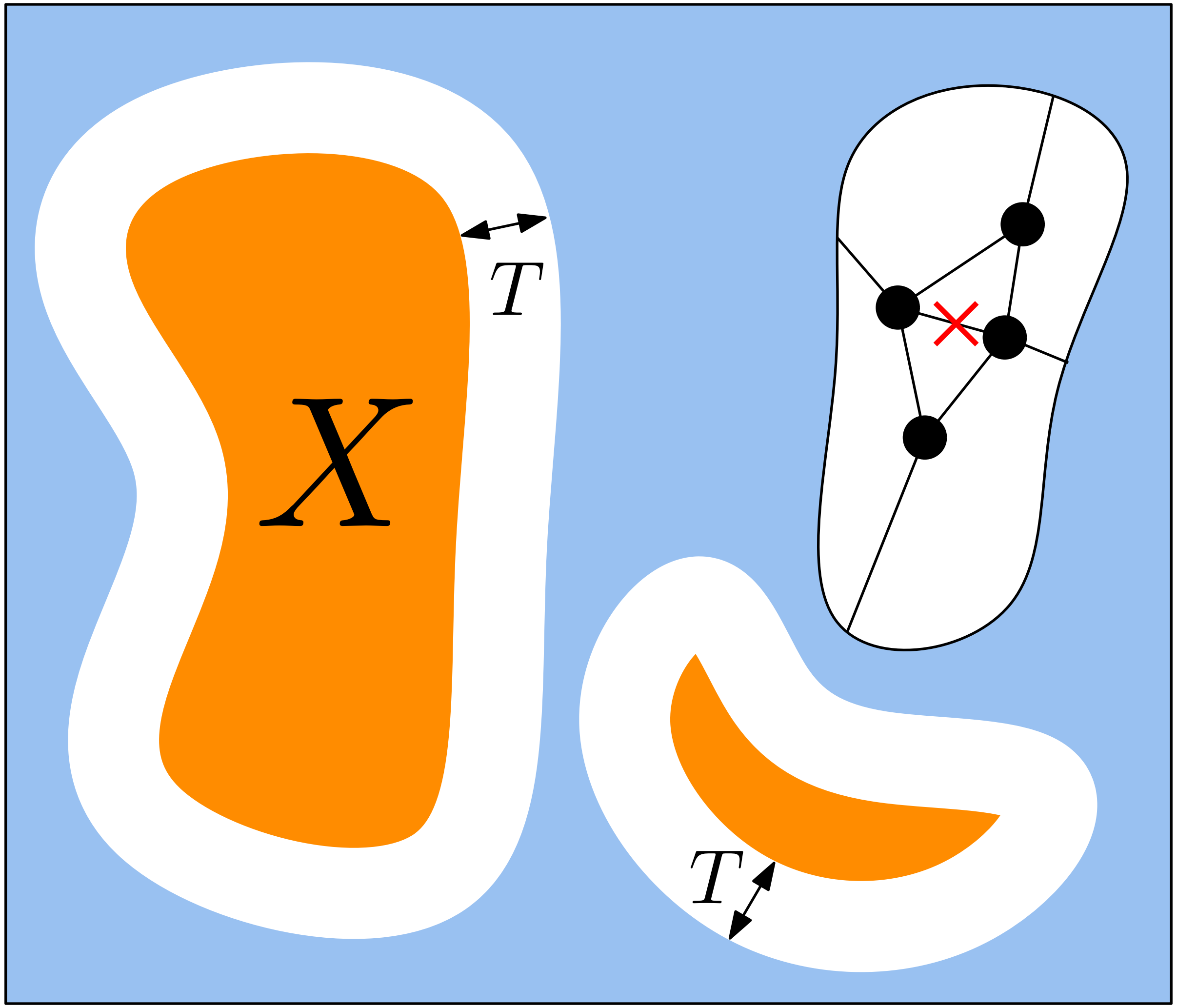

Finetely dependent distributions add restrictions not considered today

Results for Linear Programs (LPs)

Reminder:

- **Independent Set (IS)**:
  a subset of nodes $I \subseteq V$ such that $\forall v, u \in I : (v, u) \notin E$
- **Maximal Independent Set (MIS)**:
  a subset of nodes $I \subseteq V$ such that $\forall u \in V \setminus I,\ \exists v \in I : (v, u) \in E$

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Example of Distributed LP: **fractional maximum independent set**

$$\begin{aligned} \text{maximize} \sum_{v \in V} x_v & & \\ \text{subject to} \sum_{(v,u) \in E} x_v + x_u \leq 1 & & \forall v \in V \\ 0 \leq x_v \leq 1 & & \forall v \in V \end{aligned}$$

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$$

When $x_v \in \{0, 1\} \quad \forall v \in V$, the solution is a **maximum independent set**, i.e. an MIS of maximum cardinality.

Results for Linear Programs (LPs)

For LPs, LOCAL can simulate any non-signaling distribution

Example for fractional MIS

Let $T(n)$ be the locality of the non-sign. distribution

LOCAL algorithm:

- As node v, explore your radius-$T(n)$ neighborhood
- Set the output for v to be $\mathbb{E}[X_v]$

Here X_v is the rand. var. for the output of node v

Let $X = \sum_{v \in V} X_v$ be the rand. var. for the output of the entire graph

Key facts

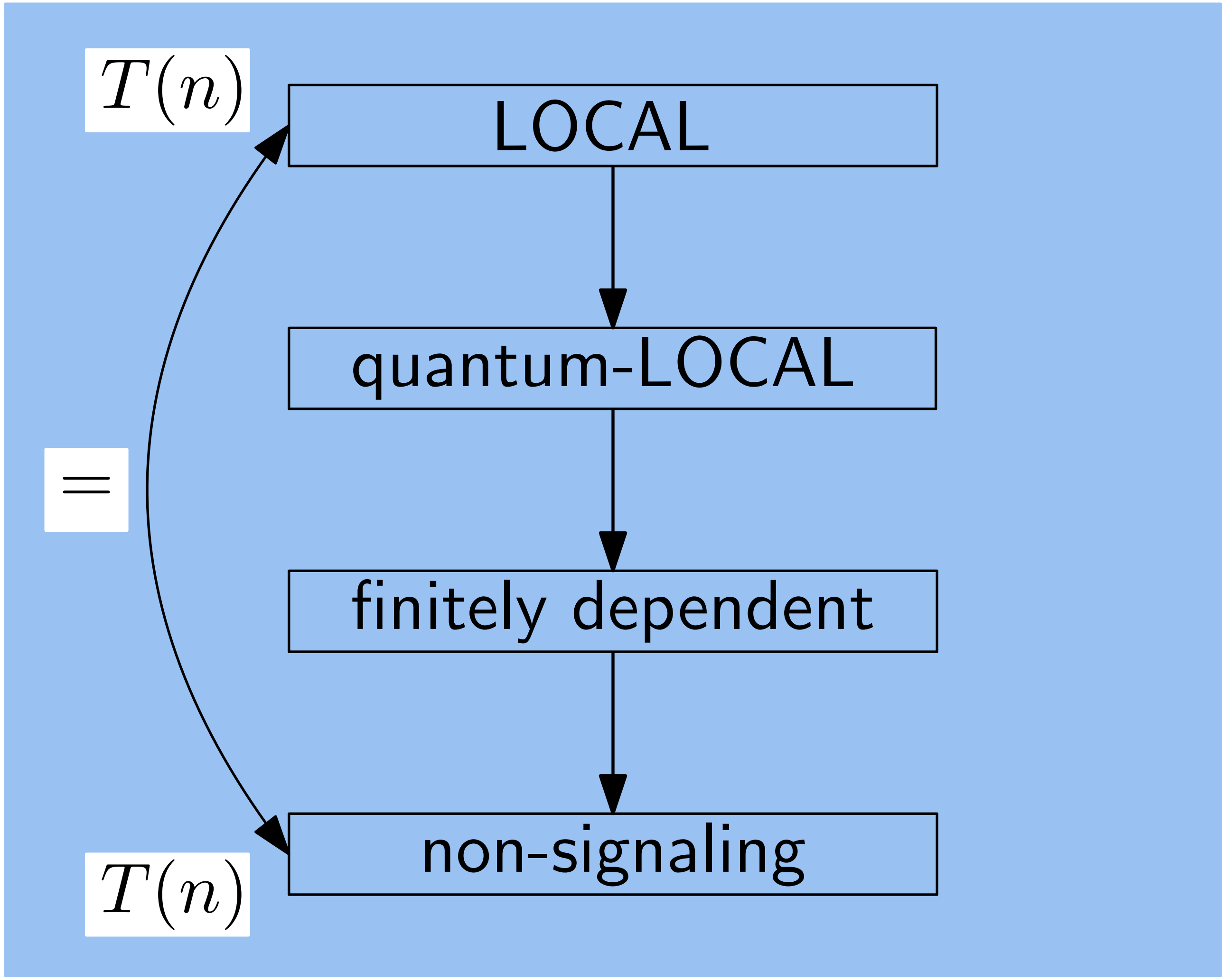
- By linearity of $\mathbb{E}$,

$$\sum_{v \in V} \mathbb{E}[X_v] = \mathbb{E}\left[\sum_{v \in V} X_v\right] = \mathbb{E}[X]$$

- By monotonicity of $\mathbb{E}$, if X is an α-approx. then $\mathbb{E}[X]$ is an α-approx.

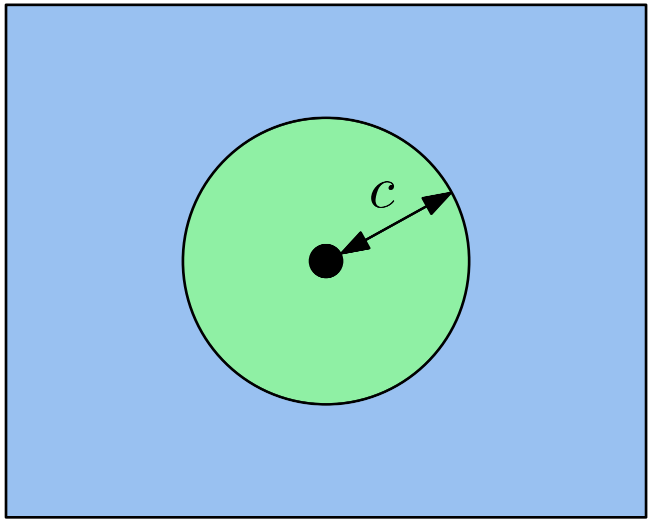
Results for Linear Programs (LPs)

Linear Programs (LPs)



Results for Locally Checable Labelly Problembs (LCLs)

LCLs: explore up to radius c to check a solution



Examples

- MIS, maximal instead of (fractional) maximum
- Coloring
- Maximal Matching
- iterated-GHZ

Reminder on GHZ

Three players A, B and C, and one referee

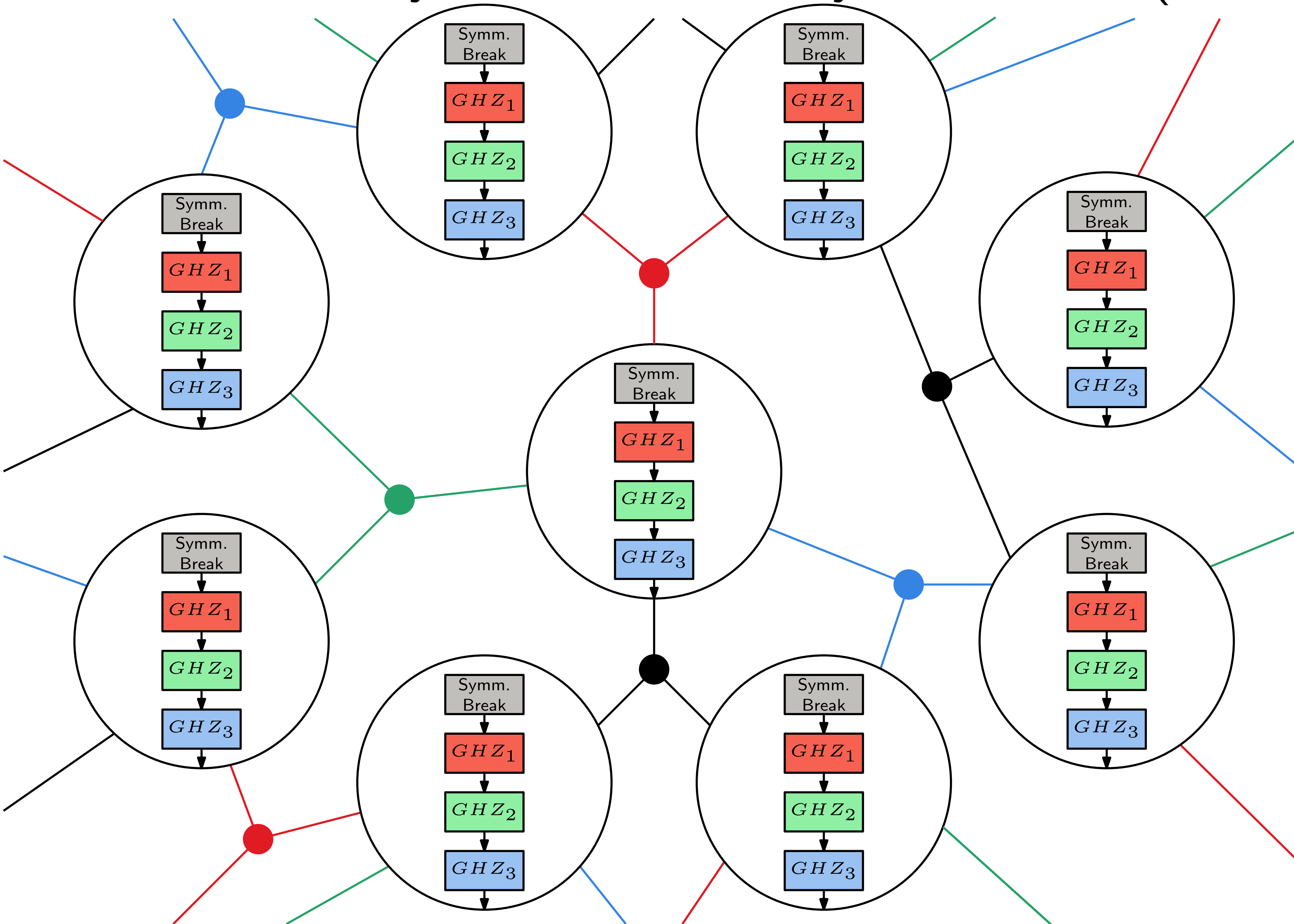
inputs x, y and z;   outputs: a, b and c

Goal: **without communicating after the input is given**, output bits such that

$$a \oplus b \oplus c = x \vee y \vee z$$

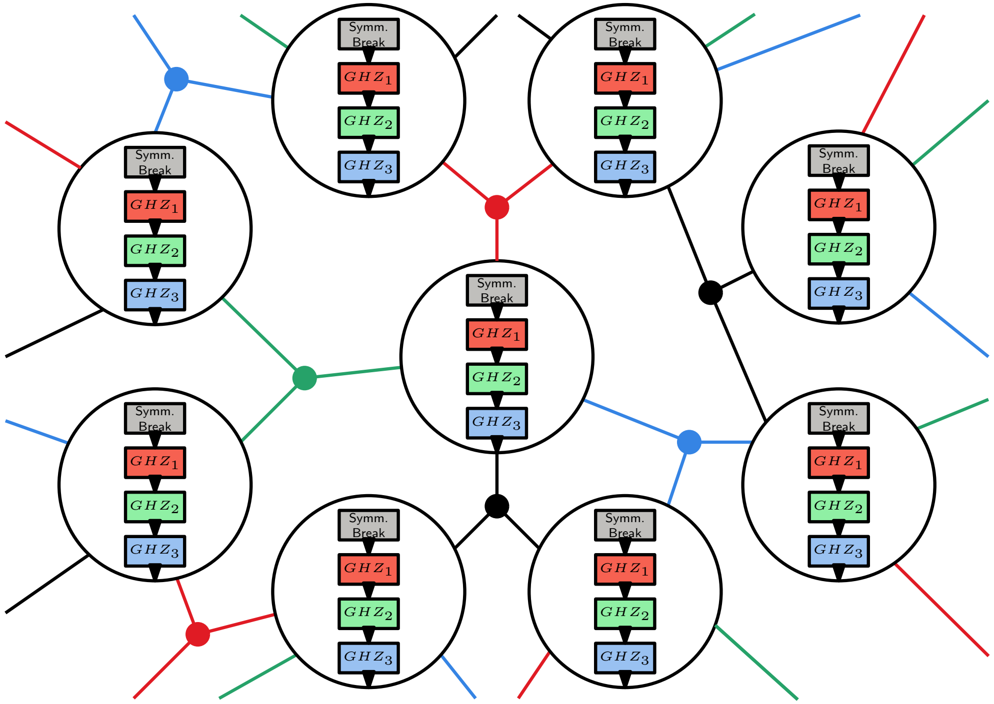
knowing that $x \vee y \vee z \in \{(0,0,0),(1,1,0),(1,0,1),(0,1,1)\}$

Results for Locally Checable Labelly Problembs (LCLs)



Results for Locally Checable Labelly Problembs (LCLs)

iterated-GHZ complexity



quantum-LOCAL: $O(1)$ rounds

- share a GHZ state
- measure

LOCAL: $O(\Delta)$ rounds

- solve symm. break. with 1^{st} neigh.
- solve GHZ_1 game with 2^{nd} neigh.
- solve GHZ_2 game with 3^{rd} neigh.
- $\ldots$
- solve GHZ_Δ game with the last neigh.